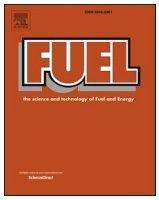
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Full Length Article



Efficient delignification of wheat straw for microbial lipid production enabled by a novel ternary deep eutectic solvent containing ethylene glycol

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ABSTRACT

Lignocellulose pretreatment often faces the problem of lignin condensation with less attention to biofuel production from total sugars, which significantly hinders biomass fractionation and energy utilization efficiency. The aim of this work was to design a novel ternary deep eutectic solvent (DES) (citric acid: choline chloride: ethylene glycol in a molar ratio of 1:1:2) for pretreatment of wheat straw with a solid/liquid ratio (S/L) of 1:20 (w/w) to achieve non-condensed lignin and high-yield fermentable sugars at 100 °C for 12 h. The introduction of ethylene glycol into ternary DES could capture the formed benzyl carbocation to protect lignin structure, inhibit excessive degradation of hemicellulose but to fermentable sugars (e.g., xylose), and improve the enzymatic digestibility of cellulose (98.73%, w/w). The mechanisms of both lignin fractionation and the in-situ lignin protection during the pretreatment of wheat straw enabled by ternary DES were systematically elucidated. Importantly, *Trichosporon cutaneum* cultivated in undetoxified DES waste liquid (hemicellulose-rich) and residue hydrolysate (cellulose-rich) could totally afford 8.7 g microbial lipid per 100 g wheat straw, which is superior to previous reports. This green in-situ lignin protection and fractionation strategy provides a new avenue for upgrading lignocellulose to microbial lipids.

1. Introduction

Microbial lipid is deemed as a superior raw material for biodiesel production, which is typically produced by oleaginous microorganisms [1–3]. Biodiesel is one of the most promising renewable fuels due to its biodegradable, environmentally friendly, and non-toxic characteristics, which acts as a feasible alternative to fossil fuels [4–7]. Nevertheless, the scale-up production of biodiesel encounters several challenges, especially high-cost carbon sources (e.g., lipids, and simple sugars), which leads to financial burdens and limits their practical applications [8]. Lignocellulosic biomass occupies the top position among the feedstocks for the production of biofuels due to its abundant storage, renewable feature, and low cost, while containing high content of carbohydrates [9,10]. Cellulose and hemicellulose are the predominant resources of fermentable sugars, accounting for 66–75% dry weight of lignocellulose [11]. Using the total sugars (cellulose and hemicellulose) of natural lignocellulosic biomass as carbon sources for the production of microbial lipid can not only enhance the application efficiency of lignocellulose but also increase the cost competitiveness of microbial lipid. However, the foremost difficulty in the efficient release of total sugars is the inherently recalcitrant structure of raw lignocellulose, which is caused by the lignocellulose structural heterogeneity and cellulose crystallinity [12,13]. Hence, efficient pretreatment approaches are vital prerequisites for biodiesel production from economically promising fermentation [14–16].

Recently, various developed pretreatment technologies (e.g., physical, chemical, and biological methods) concentrate on enhancing cellulose digestion and the production of cellulosic sugars, while neglecting lignin condensation and utilization of hemicellulosic sugars, which leads to the low utilization efficiency of total sugars [17–19]. The presence of undesired lignin condensation can not only decrease the delignification efficiency but also reduce the cellulase accessibility, since the condensed lignin redeposits on the surface of cellulose and participates in the non-productive adsorption of cellulase [20]. More importantly, hemicellulose fraction was discarded as the waste liquid in conventional biorefinery, due to the formation of strong inhibitors (e.g., furfural and 5-

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hydroxymethylfurfural) for the fermentation process [21]. At this juncture, it is crucial to develop an efficient and green pretreatment technique to prompt the liberation of fermentable sugars, reduce the formation of inhibitory components, and protect the lignin structure.

Typically, DES consists of hydrogen bond donors (HBDs) and hydrogen bond acceptors (HBAs), which are considered as green solvents due to their convenient preparation, nontoxicity, biodegradability, and biocompatibility [22]. Importantly, the structure and physicochemical properties of DESs could be governed by adjusting the constituent and distribution, hence, the interaction of DES with hemicellulose, cellulose, and lignin could be regulated [23,24]. Currently, numerous DES-mediated lignocellulose pretreatment methods focus on the delignification mechanism, cellulose enzymatic hydrolysis, and recovery of DESs, whereas overlooking the mechanism of lignin fractionation and protection, as well as the hemicellulose utilization [20,25]. DES waste liquid after lignin extraction usually contains DES constituents and hemicellulose hydrolysis products (e.g., xylose and glucose), which are often wasted without efficient application or more energy is required for detoxification before utilization. Therefore, it is desirable to develop an effective DES that can protect the lignin structure and in parallel improve the release of total sugars for the production of microbial lipids.

Wheat straw has a global production of 529 million tons per year [14], which is a sustainable and promising feedstock for producing biofuels and bioproducts [26]. Herein, an in-situ lignin protection strategy was developed by designing a novel ternary DES for efficient pretreatment of wheat straw, which could not only selectively cleave the ether or ester bonds of lignin-carbohydrate complexes (LCC) linkages and directionally protect the β-O-4 linkages of lignin, but also improve the fractionation efficiency of cellulose, hemicellulose and lignin. The mechanisms of both lignin fractionation and the in-situ lignin protection during pretreatment of wheat straw enabled by ternary DES were systematically elucidated. In addition, the ternary DES-enabled in-situ lignin protection strategy could inhibit the excessive degradation of hemicellulose but to fermentable sugars (e.g., xylose), and decrease the formation of fermentable inhibitors (e.g., furfural, 5-hydroxymethylfurfural), which increases the utilization efficiency of hemicellulose component from wheat straw. This ternary DES-enabled in-situ lignin protection strategy provides a new and economic avenue to effectively stabilize the lignin structure, sufficiently liberate the fermentable sugars, and efficiently produce microbial lipid from wheat straw.

2. Materials and methods

2.1. Materials

Wheat straw was acquired from Solarbio Science & Technology Co., Ltd. (Beijing, China). The raw feedstock was ground into 40–80 mesh particles and dried to a constant weight. Choline chloride (ChCl, ≥99.0%), citric acid (CA, ≥99.5%), acetic acid (AA, ≥99.5%), malic acid (MA, ≥99.0%), formic acid (FA, ≥98.0%), boric acid (BA, ≥99.5%), lactic acid (LA, ≥99.0%), ethylene glycol (EG, ≥99.0%) and other reagents of analytical purity grade were bought from Innochem Co., Ltd. (Beijing, China).

2.2. Deep eutectic solvent (DES) synthesis and pretreatment

Various HBDs (CA, MA, FA, AA, BA, LA and EG) and HBA (ChCl) with different molar ratios were mixed at 500 rpm to achieve a transparent and homogenous liquid, facilely furnishing the desired DES. Typically, 1.0 g dry wheat straw and 20.0 g DES were immersed in a round-bottom flask (100 mL) and heated to targeted temperatures (80–140 °C) for different times (3–24 h) and stirred at 500 rpm. Then, the obtained slurry was thoroughly washed with ethanol. Upon filtration, the resulting solid residue was separated and dried at 105 °C for 12 h in an oven. After the filtrate was evaporated to remove ethanol, the obtained lignin was precipitated by deionized water and recovered by centrifuging at 10,000 rpm for 5 min, followed by washing thoroughly with a mixture of ethanol and water (1:9, v/v). Finally, the extracted lignin was freeze-dried and stored in a desiccator, and then the resulting DES waste liquid was stored in a freezer.

2.3. Enzymatic hydrolysis

Enzymatic hydrolysis of the raw and pretreated wheat straw was evaluated at 2% (w/v) solids loading with citrate buffer (50 mM, pH 4.8), 25 FPU/g substrate cellulase (Cellic® CTec2, Filter paper activity: 195 FPU/mL, Novozyme), and 2 g/L tetracycline hydrochloride solution. The process was performed in an Erlenmeyer flask at 50 °C for 72 h at 150 rpm in a shaker. The hydrolysate was collected periodically, and finally the enzyme was inactivated by incubating at 100 °C for 10 min [27,28]. The glucose yield was determined following the below equation:

$$\text{Glucose yield } (\%) = \frac{\text{Glucose in the enzymatic liquid (g)}}{\text{Glucose in the substrate (g)}} \times 100\% \quad (1)$$

2.4. Strain and cultivation

Trichosporon cutaneum ACCC 20119 (*T. cutaneum*), provided by ACCC (Agricultural culture collection of China), was cultured on YPD (yeast peptone dextrose) medium (yeast extract: 10 g/L, tryptone: 20 g/L, glucose: 20 g/L, and agar: 20 g/L) and stored at 4 °C. The medium was sterilized at 115 °C for 15 min using an autoclave. To achieve the initial dry cell weight (DCW) of 5 g/L, a suitable amount of preculture was added into the fermentation medium [21].

2.5. Batch fermentation

The lipid production by *T. cutaneum* was estimated from the diluted DES waste liquid and solid residue hydrolysates via batch fermentation for 144 and 72 h, respectively [29]. The batch culture was carried out in an Erlenmeyer flask at 28 °C and 180 rpm in a rotary shaker. The initial pH of the culture was set at 6.0. The component of the fermentation medium was as follows: yeast extract 1.0 g/L, $MgSO_4{\cdot}7H_2O$ 0.5 g/L, KH_2PO_4 1.0 g/L, $(NH_4)_2SO_4$ 2.2 g/L, and the sugars of the mediums were obtained from DES waste liquids and solid residue hydrolysates. In addition, 10% (v/v) of seed culture medium was utilized as inoculum for lipid production. To supervise the yeast growth, samples were extracted every 24 or 12 h, centrifuged at 10,000 rpm for 2 min, washed, and freeze-dried for lipid extraction.

2.6. Lipid extraction

Lipid extraction was carried out based on the recognized method in the published literature [29,30]. Typically, 0.4 g freeze-dried cells were suspended in 4 M HCl solution for 1 h, ultrasonically extracted at 65 °C for 45 min, boiled for 10 min, and cooled rapidly in an ice water bath to facilitate the cell membrane lysis. To the suspension solution, chloroform/methanol (2:1, v/v) was added. Then, the obtained solution was incubated at ambient temperature (25 ± 1 °C) for 2 h, followed by centrifuging at 10,000 rpm for 2 min. The organic phase was evaporated at 80 °C for 4 h to remove chloroform, methanol, and water, and oven-dried to a constant weight to achieve lipid. DCW, lipid yield, and lipid content were determined following equations (2), (3), and (4).

$$\text{DCW (g/L)} = \frac{\text{Weight of dried yeast cell}}{\text{Volume of fermentation medium}} \quad (2)$$

$$\text{Lipid yield (g/L)} = \frac{\text{Weight of extracted lipid}}{\text{Volume of fermentation medium}} \quad (3)$$

$$\text{Lipid content}\ (\%) = \frac{\text{Weight of extracted lipid}}{\text{Weight of dried yeast cell}} \times 100\% \quad (4)$$

2.7. Fatty acid methyl esters (FAMEs) composition determination

Transesterification was performed to determine the FAMEs composition via an acid catalyst utilizing the methanol-sulfuric acid method [30]. Typically, the dried lipid and methanol-sulfuric acid (2%) mixture were added to a pressure tube, then the obtained solution was heated at 70 °C for 4 h and washed with deionized water to maintain the neutral pH. Subsequently, the addition of ethyl acetate to the mixture and the fatty acid was obtained from the organic phase. Finally, the fatty acid composition was identified based on the SUPELCO standard mix. The composition of lipid was extracted with chloroform and analyzed by GC–MS (Agilent 6890 N GC/5973 MS) and GC (Agilent 7890B) equipped with an HP-5 column (30 m × 0.320 mm × 0.25 μm) and an FID (flame ionization detector).

2.8. Characterization of deep eutectic solvents (DESs), cellulose-rich residues, and extracted lignin

2.8.1. Characterization of deep eutectic solvents (DESs)

The Kamlet-Taft (K-T) solvatochromic parameters (α, β, and π*) of DES were measured by the published method [31]. The detailed process of parameters was listed in Section S1. Fourier transform infrared spectroscopy (FT-IR) spectra of DESs were determined on a Nicolet IS50 AEM (Thermo Fisher) with a KBr disk in a wavenumber range from 4000 to 400 cm^{-1}.

2.8.2. Characterization of cellulose-rich residues

Chemical compositions of raw wheat straw and cellulose-rich residues were estimated by NREL (National Renewable Energy Laboratory) standard technique [32]. The concentrations of sugars and inhibitors were detected by HPLC (Agilent 1260, Hi-Plex H, 300 × 7.7 mm, 8 μm) equipped with a differential refractive index detector (RID) with 0.05 mol/L dilute H_2SO_4 as the mobile phase (0.5 mL/min). The temperature of the column and RID was kept at 50 °C. The morphology changes of raw and pretreated wheat straw were analyzed by scanning electron microscope (SEM, ZEISS SIGMA300). N_2 adsorption–desorption isotherms of cellulose-rich residues were performed on a Micromeritics ASAP 2460 analyzer at − 196 °C to measure Brunauer-Emmett-Teller (BET) surface areas. FT-IR spectra of cellulose-rich residues were determined on a Nicolet IS50 AEM (Thermo Fisher) with a KBr disk in a wavenumber range from 4000 to 400 cm^{-1}.

X-ray diffraction (XRD) patterns (2θ = 10–90°) of cellulose-rich residues were examined on a Bruker D8 Advance X-ray powder diffractometer with Cu-Kα (λ = 1.542 Å). The crystallinity index (CrI) of raw wheat straw and residues was calculated by the reported method. The CrI value was examined following the equation:

$$\text{CrI}(\%) = \frac{(\text{I002} - \text{Iam})}{\text{I002}} \times 100\% \quad (5)$$

where I_{002} is the intensity of the signal at around 22.2°; I_{am} is the intensity of the amorphous region at around 18.2° [33].

Cellulose hydrophobicity and accessibility of both treated and untreated wheat straw were measured utilizing the Rose Bengal reagent dyeing method [34] and Direct Red dye (DR28) adsorption [35], respectively. The detailed processes of hydrophobicity and accessibility were summarized in Sections S2 and S3, respectively.

2.8.3. Characterization of extracted lignin

The molecular weight values of extracted lignin were detected by GPC (gel permeation chromatography) (Agilent 1260, PLgel MIXED-D, 300 × 7.5 mm, 5 μm) equipped with RID [36]. The detailed process was listed in Section S4. The purity of extracted lignin was estimated by the NREL method [32]. Moreover, ^1H and ^{13}C NMR spectra of extracted lignin were measured using a Bruker Biospin AG-400 spectrometer. FT-IR spectra of extracted lignin were determined on a Nicolet IS50 AEM (Thermo Fisher) with a KBr disk in a wavenumber range from 4000 to 400 cm^{-1}.

2.9. Statistical analysis

All the experiments were performed in triplicates. All data reported in the study are the mean values with standard deviations of <5%.

2.10. Experimental design

An effective and green process for the transformation of wheat straw into microbial lipids enabled by ternary DES pretreatment was illustrated in Fig. 1.

3. Results and discussion

3.1. Influence of different hydrogen bond donors on the deep eutectic solvent-mediated biomass pretreatment

3.1.1. Changes of chemical compositions in wheat straw

The changes of chemical compositions in wheat straw after pretreatment could be conducted to evaluate the treatment effects. Numerous parameters such as hemicellulose removal, cellulose recovery, delignification, solid yield, components of DES solution and cellulose-rich residues should be comprehensively analyzed. Specially, the pretreatment temperature and DES acidity were the sensitive parameters for the pretreatment process, while cellulose recovery and lignin removal were the sensitive parameters for enzymatic digestion. The content of fermentable sugars (glucose and xylose), and the concentration of inhibitors (furfural and 5-hydroxymethylfurfural) were the sensitive parameters for the fermentation process [37].

Wheat straw was pretreated with different HBDs in the ChCl-based DESs at 100 °C for 12 h in S/L of 1:20 (w/w), and the pretreated hydrolysates and solid residues were isolated (Table 1). Overall, Table 1 illustrates that the solid yield was strongly affected by the removal of hemicellulose and lignin [23]. The binary DESs, including EG: ChCl = 2:1 and CA: ChCl = 1:1, revealed high solid yield (78.54% and 72.87%), low delignification (35.67% and 41.52%), and low hemicellulose removal (26.25% and 33.67%), respectively. However, the little difference in the change of cellulose components (cellulose recovery > 87.53%) was observed due to its rigid structure, which is in agreement with the results reported by Lei et al. [27]. The inferior performance without CA in EG: ChCl = 2:1 possibility indicated that the absence of sufficient acidity cannot facilitate the hemicellulose and lignin degradation, while the severe lignin condensation and high viscosity were observed without EG in CA: ChCl = 1:1 [12]. This result is in line with the previous reports [38], where the absence of maleic acid could impede biomass degradation.

To further improve the pretreatment efficiency, a series of novel ternary DESs were designed by adjusting the molar ratio of CA, ChCl, and EG. In detail, the solid yield first decreased from 62.32% to 45.15% then increased to 58.58% and the delignification first enhanced from 70.45% to 92.37% then decreased to 68.95% when the molar ratio changed from 5:1:2 to 1:1:8. It was found that the acidity and viscosity of DES could remarkably influence the efficiency of pretreatment, which are considered as the sensitive parameters for biomass pretreatment [37]. In the case of CA: ChCl: EG = 5:1:2 that had relatively high acidity (high content of CA), but a low lignin fractionation was obtained, which was attributed to its excessive viscosity obstructing mass transfer during pretreatment [12]. However, CA: ChCl: EG = 1:1:8 possessed low viscosity (high content of EG) but still low lignin removal, primarily due to its insufficient acidity [39]. Gratifyingly, CA: ChCl: EG = 1:1:2 could afford the low solid yield (45.15%), high delignification (92.37%) and

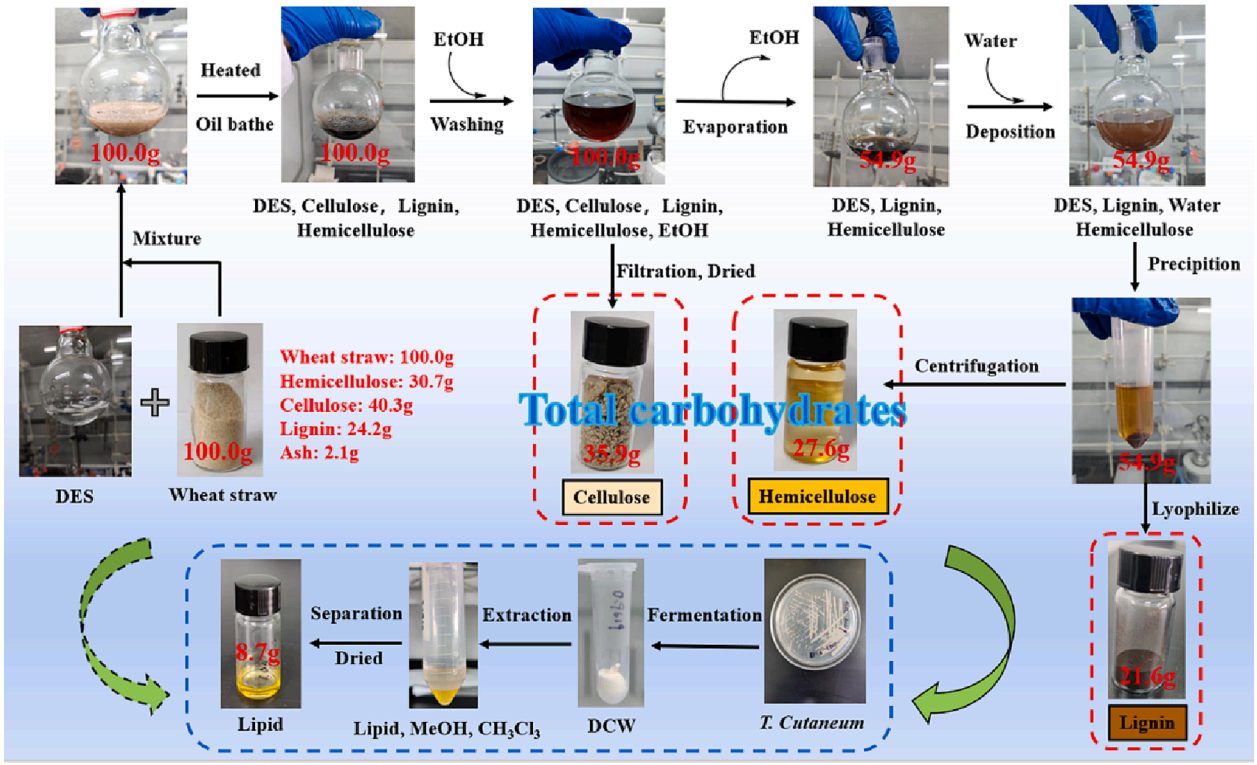


Fig. 1. Schematic illustration of wheat straw pretreatment and fermentation enabled by the developed DES.

Table 1
Influence of various DES compositions on pretreatment of wheat straw.[a]

NO.	DES	Solid yield (%)	C recovery (%)	H removal (%)	L removal (%)	Sugar analysis (g/L)[b]		Composition analysis (%)[c]			Inhibitor (g/L)[b]	
						Glucose	Xylose	Cellulose	Hemicellulose	Lignin	FF	HMF
Raw	–	–	–	–	–	–	–	40.35 ± 1.67	30.52 ± 1.74	24.16 ± 1.56	–	–
1	EG: ChCl = 2:1	78.54 ± 0.83	94.26 ± 1.21	26.25 ± 1.42	35.67 ± 1.36	0.69 ± 0.06	2.85 ± 0.26	46.47 ± 1.21	25.83 ± 1.04	19.71 ± 1.52	0.13 ± 0.01	0.08 ± 0.02
2	CA: ChCl = 1:1	72.87 ± 1.36	91.37 ± 2.13	33.67 ± 3.12	41.52 ± 1.25	0.95 ± 0.11	6.23 ± 0.75	58.36 ± 1.59	18.61 ± 2.02	15.23 ± 1.85	0.16 ± 0.02	0.12 ± 0.01
3	CA: ChCl: EG = 5:1:2	62.32 ± 1.28	92.41 ± 1.42	76.54 ± 2.56	70.45 ± 1.43	1.08 ± 0.01	6.72 ± 1.26	61.91 ± 1.99	15.08 ± 1.38	14.02 ± 0.82	0.04 ± 0.01	0.02 ± 0.02
4	CA: ChCl: EG = 2:1:2	49.72 ± 0.98	91.55 ± 2.62	84.16 ± 2.16	87.66 ± 2.22	1.99 ± 0.15	9.07 ± 0.63	81.49 ± 1.01	6.36 ± 0.27	7.14 ± 1.18	0.03 ± 0.01	0.01 ± 0.00
5	CA: ChCl: EG = 1:1:2	45.15 ± 0.15	89.18 ± 1.48	90.35 ± 1.32	92.37 ± 1.26	2.38 ± 0.03	10.55 ± 1.11	89.12 ± 1.07	3.04 ± 0.11	5.71 ± 0.12	trace	trace
6	CA: ChCl: EG = 1:2:2	48.83 ± 1.42	90.62 ± 2.31	87.03 ± 1.73	90.14 ± 1.67	2.05 ± 0.12	9.51 ± 1.57	84.88 ± 2.23	4.02 ± 0.65	6.01 ± 0.96	0.02 ± 0.00	trace
7	CA: ChCl: EG = 1:5:2	51.12 ± 1.18	88.44 ± 1.78	84.22 ± 2.21	85.64 ± 1.48	1.89 ± 0.15	8.36 ± 1.62	80.49 ± 2.34	5.91 ± 0.83	8.58 ± 1.36	0.03 ± 0.00	trace
8	CA: ChCl: EG = 1:1:4	55.47 ± 1.51	89.52 ± 2.09	80.54 ± 1.46	80.57 ± 1.33	1.43 ± 0.08	6.75 ± 1.72	75.66 ± 2.03	8.59 ± 1.08	9.75 ± 1.32	0.02 ± 0.00	0.01 ± 0.00
9	CA: ChCl: EG = 1:1:8	58.58 ± 1.26	87.76 ± 1.98	78.66 ± 2.31	68.95 ± 2.35	1.37 ± 0.11	7.31 ± 0.92	69.55 ± 1.32	9.94 ± 1.07	11.95 ± 1.36	0.03 ± 0.00	0.01 ± 0.00
10	FA: ChCl: EG = 1:1:2	50.36 ± 2.23	89.34 ± 2.51	86.66 ± 2.31	85.58 ± 1.52	1.71 ± 0.08	7.71 ± 1.38	82.42 ± 1.98	6.24 ± 0.87	7.75 ± 1.03	0.03 ± 0.00	trace
11	AA: ChCl: EG = 1:1:2	58.13 ± 1.61	88.72 ± 1.68	75.35 ± 2.18	77.82 ± 1.63	1.25 ± 0.15	7.19 ± 0.51	73.75 ± 1.67	8.73 ± 0.91	10.02 ± 1.05	0.04 ± 0.00	0.01 ± 0.01
12	LA: ChCl: EG = 1:1:2	53.81 ± 2.31	87.53 ± 1.45	83.57 ± 1.22	82.75 ± 2.79	1.54 ± 0.24	7.55 ± 1.33	78.68 ± 1.08	7.11 ± 1.06	8.04 ± 0.83	0.03 ± 0.00	0.02 ± 0.01
13	MA: ChCl: EG = 1:1:2	46.56 ± 1.44	88.78 ± 2.41	89.27 ± 2.12	90.83 ± 1.96	2.01 ± 0.11	9.43 ± 1.77	85.76 ± 2.15	4.92 ± 0.45	5.82 ± 0.91	0.02 ± 0.01	trace
14	BA: ChCl: EG = 1:1:2	61.52 ± 2.23	90.58 ± 1.66	70.96 ± 1.97	55.13 ± 2.48	0.91 ± 0.22	6.71 ± 0.64	71.08 ± 1.41	9.97 ± 1.03	10.86 ± 0.45	0.03 ± 0.01	0.01 ± 0.00

[a]Reaction conditions: S/L = 1:20, 100 °C, 12 h.
[b]The components of DES waste liquid.
[c]The components of cellulose-rich residues.
C recovery = Cellulose recovery, H removal = Hemicellulose removal, L removal = Lignin removal, HMF = 5-hydroxymethylfurfural, FF = furfural.

hemicellulose removal (90.35%), and high cellulose yield (89.12%) for the pretreatment of wheat straw. Compared with the reported ternary DES [40], CA: ChCl: EG = 1:1:2 exhibited a relatively lower solid yield (45.15% *vs* 60.88%) and higher delignification rate (92.37% *vs* 71.64%). These results manifested that a suitable content of CA in DES could efficiently destroy the ether or ester bonds of LCC linkages, causing the sufficient removal of hemicellulose and lignin [41]. On the other hand, an appropriate amount of EG greatly inhibited lignin condensation and reduced DES viscosity, leading to a relatively high fractionation efficiency [40,42].

Viewing the importance of the acidity of DES for efficient pretreatment of wheat straw, the pKa value of HBDs in DESs was compared (Table 1), and found to be negatively correlated with the lignin fractionation [36]. In this work, the delignification (lignin removal) after pretreatment of wheat straw using different DESs bearing various types of HBDs was presented in the order of CA: ChCl: EG (92.37%) > MA: ChCl: EG (90.83%) > FA: ChCl: EG (85.58%) > LA: ChCl: EG (82.75%) > AA: ChCl: EG (77.82%) > BA: ChCl: EG (69.13%). A lower value of pKa in HBDs (CA: 3.13 < MA: 3.40 < FA: 3.75 < LA: 3.86 < AA: 4.74 < BA: 8.91) corresponding to the higher acidity could supply more active protons to attack and break the ether or ester bonds of LCC linkages, leading to higher delignification [22,43]. The relatively lower value of pKa in HBD was beneficial to the degradation of lignin and hemicellulose, which was also illustrated by Mankar et al. [43]. Also, three –COOH groups in CA provided more acidity for lignin fractionation. The results further indicated the superior pretreatment efficiency of CA: ChCl: EG = 1:1:2 to those of other DESs, which was closely correlated with its relatively high acidity and moderate viscosity.

In addition, the influence of temperature (80–140 °C), time (3–24 h), and S/L (1:10–1:25, w/w) on the pretreatment of wheat straw was studied (Table S1). The increment of temperature (80–140 °C) promoted the hemicellulose and lignin removal, resulting in 43.27–65.55% solid yield and 63.06–95.81% delignification, however, the cellulose yield decreased from 89.12% to 84.12% when the temperature increased from 100 °C to 140 °C. Meanwhile, with the increase of the reaction time from 3 to 24 h, the solid yield decreased from 60.18% to 44.95%, while the cellulose yield first enhanced from 66.14% (3 h) to 89.12% (12 h) and then reduced to 85.03% (24 h). The variation trend of composition plotted with temperature and time is consistent with previous investigations [27]. Too high a temperature or excessively prolonged time caused the increased loss of cellulose and the formation of inhibitors, which greatly compromised the subsequent enzymatic hydrolysis [44]. When the S/L was decreased from 1:10 to 1:25, the solid yield decreased from 57.26% to 44.24% and delignification increased from 85.35% to 93.15%, but the cellulose yield increased from 84.07% to 89.12% then decreased to 85.24%. It was indicated that a relatively high ratio of S/L (1:10) caused the insufficient removal of hemicellulose and lignin, while a low ratio of S/L (1:25) led to excessive removal of lignin, loss of cellulose, and formation of inhibitors [28]. In view of protecting the lignin structure and improving the release of total sugars, the optimal S/L was selected at 1:20 at the pretreatment temperature of 100 °C for 12 h.

3.1.2. Degradation of carbohydrates in wheat straw after delignification

The content of obtained total sugars would be diminished by the degradation of carbohydrates (hemicellulose and cellulose) from wheat straw. In a general biomass pretreatment process, polysaccharides are degraded into monosaccharides, followed by further dehydration to furfural and 5-hydroxymethylfurfural in relatively high quantities, which are deemed as inhibitors in the fermentation process [27]. To study the efficiency of DES-mediated pretreatment of wheat straw on the degradation of hemicellulose and cellulose, the concentrations of glucose, xylose, and inhibitors in the ternary DES waste liquid were determined (Table 1). Large amounts of monosaccharides (e.g., xylose: 2.85–10.55 g/L, and glucose: 0.69–2.38 g/L) but much lower amounts of inhibitors (e.g., furfural: < 0.16 g/L and 5-hydroxymethylfurfural: < 0.12 g/L) were detected. Also, the content of glucose or 5-hydroxymethylfurfural was lower than that of xylose or furfural, implying that the degradation of cellulose was more difficult than that of hemicellulose [27]. Especially, the wheat straw pretreated by CA: ChCl: EG = 1:1:2 released a higher content of xylose (10.55 g/L) and glucose (2.38 g/L), but trace amounts of inhibitors (≪ 0.01 g/L) than those of EG: ChCl = 2:1 (xylose: 2.85 g/L, and glucose: 0.69 g/L) and CA: ChCl = 1:1 (xylose: 6.23 g/L, and glucose: 0.95 g/L) pretreatments. This could explain why a suitable content of CA in DES improved the hemicellulose removal and an appropriate amount of EG protected the lignin structure and stabilized xylose to impede the production of inhibitors and pseudo-lignin [19]. The optimized ternary DES (CA: ChCl: EG = 1:1:2) pretreatment of wheat straw exhibited the highest content of monosaccharides and trace inhibitors, which was beneficial to the subsequent enzymatic hydrolysis and lipid production.

3.2. Characterization of cellulose-rich residues

The efficiency of enzymatic hydrolysis of the DES pretreatment of wheat straw was determined by the structural changes of cellulose-rich residues such as morphological changes, surface area, crystallinity, functional group, hydrophobicity, and accessibility [45]. At this juncture, SEM, FT-IR, BET, XRD, Rose Bengal dye, and Direct Red 28 dye methods were utilized to analyze the structural changes in the extracted cellulose.

3.2.1. Morphological analysis of cellulose-rich residues

SEM technology was utilized to observe the morphological changes of cellulose after DES pretreatment. SEM images display that the surface morphology of raw wheat straw was dense and smooth (Fig. 2a), while EG: ChCl = 2:1 (Fig. 2b) and CA: ChCl = 1:1 (Fig. 2c) pretreated wheat straw slightly damaged the cellulose structure. In contrast, the surface morphology of wheat straw pretreated by CA: ChCl: EG = 1:1:2 (Fig. 2d) was looser and rougher, which was decomposed into independent fiber bundles, forming numerous pores on the fiber surface. The efficient C-O bond cleavage of LCC linkages led to the removal of large amounts of hemicellulose and lignin [23,43], which was in line with the low solid yield and high delignification (Table 1). The change of morphology between untreated and pretreated wheat straw is in agreement with the previous literature [43]. The severe breakdown of fibers in cellulose would be helpful to boost cellulase accessibility and promote enzymatic hydrolysis.

3.2.2. Brunauer Emmett Teller surface area analysis of cellulose-rich residues

As summarized in Table 1, the BET surface areas of cellulose-rich residues after pretreatment with different DESs were higher (1.18–4.89 m^2/g) than that of raw wheat straw (0.95 m^2/g) (Fig. S1) elevated by about 5.1, 1.2, and 1.5 folds in the case of CA: ChCl: EG = 1:1:2, EG: ChCl = 2:1, and CA: ChCl = 1:1, respectively. These enhanced surface areas of cellulose components could be ascribed to the degradation of hemicellulose and lignin [45], which was in agreement with the sufficient deconstruction of the cellulose surface structure illustrated by SEM (Fig. 2). Considering that the increased surface areas could improve cellulase accessibility for achieving high enzymatic digestibility, the wheat straw pretreated with CA: ChCl: EG = 1:1:2 would be easier to undergo enzymolysis than that with binary DESs. A similar change of the BET surface area after pretreatment was observed using a reported ternary DES [22].

3.2.3. Crystalline structure analysis of cellulose-rich residues

The crystalline properties of the original wheat straw and cellulose-rich residues were determined by the XRD technique (Table 2 and Fig. S2). The diffraction patterns of cellulose-rich residues revealed similar characteristic signals to that of the untreated wheat straw, proving no change of crystalline cellulose after DES pretreatment [36]. The crystallinity index (CrI) has been utilized to assess the efficiency of

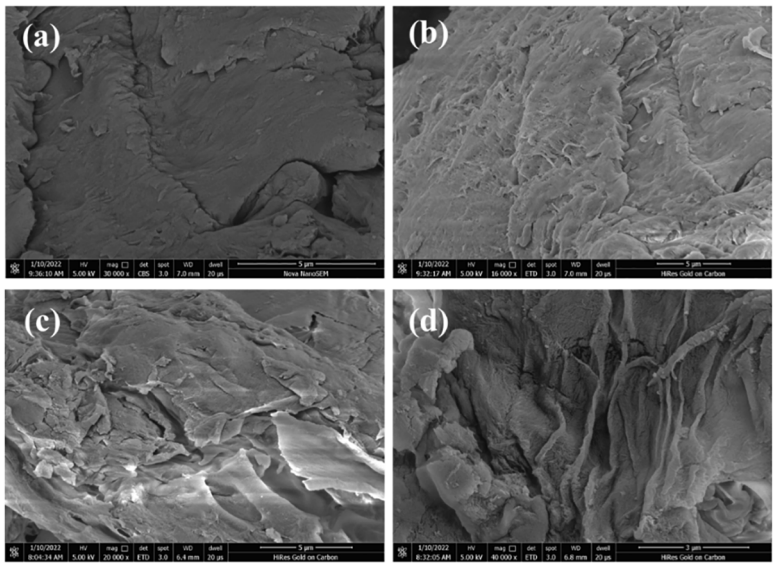


Fig. 2. SEM images of raw wheat straw (a) and wheat straw pretreated with various DESs: EG: ChCl = 2:1 (b), CA: ChCl = 1:1 (c), and CA: ChCl: EG = 1:1:2 (d).

Table 2
Kamlet-Taft parameters of different DESs, and the hydrophobicity, accessibility, BET surface area, and crystallinity index (CrI) of cellulose.

NO.	DES	α	β	π*	α-β	Hydrophobicity (mL/g substrate)	Cellulose accessibility (mg/g substrate)	BET surface area (m^2/g)	CrI (%)
Raw	–	–	–	–	–	2.67 ± 0.11	98.65 ± 1.03	0.95 ± 0.02	32.57 ± 0.34
1	EG: ChCl = 2:1	0.89	0.75	0.81	0.14	2.12 ± 0.08	105.83 ± 1.34	1.18 ± 0.06	41.68 ± 1.36
2	CA: ChCl = 1:1	1.01	0.69	0.89	0.32	1.95 ± 0.04	140.26 ± 1.07	1.47 ± 0.12	52.46 ± 0.54
3	CA: ChCl: EG = 5: 1: 2	1.48	0.76	0.95	0.72	1.87 ± 0.08	165.06 ± 1.24	2.54 ± 0.03	57.52 ± 0.21
4	CA: ChCl: EG = 2: 1: 2	1.40	0.66	1.03	0.74	0.36 ± 0.01	248.12 ± 0.98	3.36 ± 0.01	61.05 ± 0.75
5	CA: ChCl: EG = 1: 1: 2	1.36	0.58	1.19	0.78	0.07 ± 0.01	393.17 ± 1.31	4.89 ± 0.35	69.24 ± 0.28
6	CA: ChCl: EG = 1: 2: 2	1.32	0.57	1.17	0.75	0.62 ± 0.02	328.58 ± 2.04	4.12 ± 0.11	67.19 ± 0.16
7	CA: ChCl: EG = 1: 5: 2	1.21	0.52	1.06	0.69	1.19 ± 0.05	262.78 ± 1.52	3.33 ± 0.27	65.54 ± 0.24
8	CA: ChCl: EG = 1: 1: 4	1.12	0.57	1.01	0.55	1.23 ± 0.04	213.45 ± 1.76	3.05 ± 0.18	59.66 ± 0.33
9	CA: ChCl: EG = 1: 1: 8	1.04	0.63	0.94	0.41	1.35 ± 0.06	184.52 ± 1.45	2.13 ± 0.09	57.19 ± 0.19
10	FA: ChCl: EG = 1: 1: 2	1.27	0.59	1.07	0.68	1.25 ± 0.02	265.13 ± 2.09	3.26 ± 0.19	63.22 ± 0.25
11	AA: ChCl: EG = 1: 1: 2	1.19	0.62	0.96	0.57	1.28 ± 0.03	195.42 ± 1.83	2.51 ± 0.14	57.44 ± 1.26
12	LA: ChCl: EG = 1: 1: 2	1.21	0.59	1.02	0.62	1.39 ± 0.05	232.41 ± 2.36	3.17 ± 0.23	59.73 ± 0.72
13	MA: ChCl: EG = 1: 1: 2	1.31	0.55	1.15	0.76	0.09 ± 0.01	287.39 ± 1.32	3.98 ± 0.11	66.06 ± 1.17
14	BA: ChCl: EG = 1: 1: 2	1.06	0.61	0.92	0.45	1.69 ± 0.12	173.23 ± 1.61	1.76 ± 0.08	55.25 ± 1.22

DES pretreatment [44]. The CrI value of raw wheat straw was 32.57%, whereas that of all pretreated residues increased (41.68–69.24%), in which the ternary DESs showed superior performance to the binary counterparts, consistent with the results reported in previous literature [38,45]. The efficient removal of hemicellulose, lignin, and non-crystallinity composition resulted in enhancing the relative crystallinity of cellulose [27]. Among the prepared DESs, the CrI of CA: ChCl: EG = 1:1:2 (69.24%) was higher than that of the DESs bearing different HBDs, which was mainly attributed to different acidity (pKa) of HBDs (see section 3.1.1). In addition, the CrI values of cellulose-rich residues first increased from 57.52% to 69.24% and then decreased to 57.19% as the molar ratio of CA/ChCl/EG in the ternary DESs changed from 5:1:2 to 1:1:8, further demonstrating that the suitable acidity and viscosity of CA: ChCl: EG = 1:1:2 was helpful to the efficient removal of hemicellulose and lignin, closely correlated with excellent enzymatic hydrolysis.

3.2.4. *Fourier transform infrared spectra analysis of cellulose-rich residues*

The changes in the functional group of residues after pretreatment with various DESs were determined by FT-IR. The peak at 1730 cm^{-1} belonging to the C═O in hemicellulose component became weaker clearly with the changes in the molar ratio of CA/ChCl/EG (Fig. S3a) and HBDs type of DESs (Fig. S3b) in comparison with the raw wheat straw,

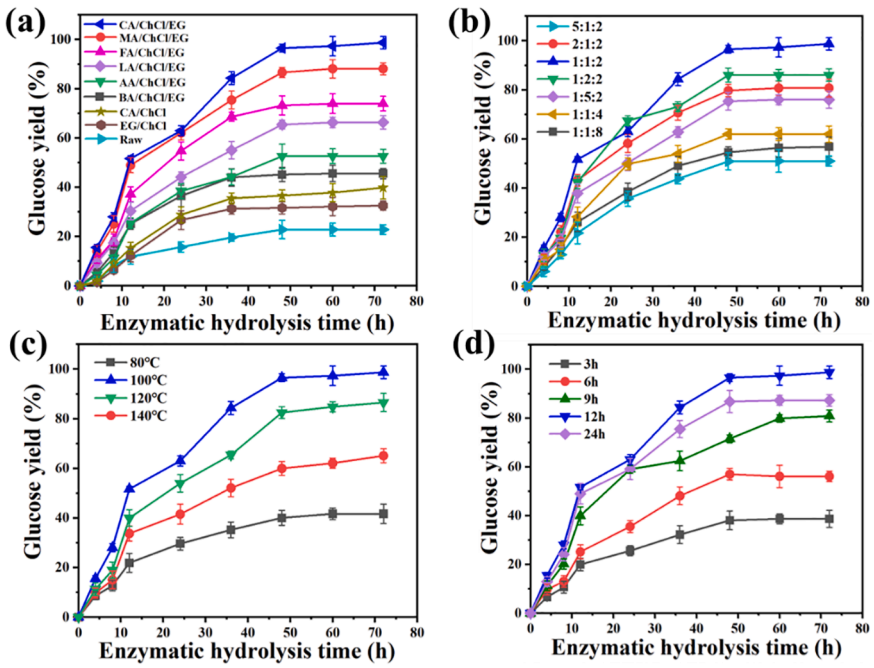


Fig. 3. Enzymatic hydrolysis of raw and pretreated wheat straw using different DESs.

indicating that hemicellulose was removed efficiently after pretreatment by these DESs [46]. The signals at around 1510 cm^{-1} and 1240 cm^{-1} contributed to aromatic C-C stretching and C-O vibration in lignin composition, which decreased or even disappeared because of lignin removal [36]. Interestingly, the bonds at 900 and 1103 cm^{-1} were assigned to the C-H vibration and C-O stretching in crystalline cellulose of all the pretreated feedstocks became relatively stronger than that of raw wheat straw, reflecting the overall enhancement in cellulose crystallinity [46], which was also observed by XRD results (Fig. S2). The intensity of signals at 1035, 1056, 1371, and 2896 cm^{-1} corresponded to the C-O and C-H bending in cellulose, which had an increased intensity after DESs pretreatment. The results indicated that cellulose was enriched in the pretreatment of wheat straw by DESs [27]. The outstanding pretreatment efficiency of CA: ChCl: EG = 1:1:2 could be due to the almost complete removal of hemicellulose and lignin (Table 2).

3.2.5. Hydrophobicity analysis of cellulose-rich residues

The surface hydrophobicity of solid cellulose-rich residues after DESs pretreatment was determined by Rose Bengal dye (Table 2). The hydrophobicity of DES-pretreated samples was decreased by 0.55–2.60 mL/g substrate in comparison with that of raw wheat straw (2.67 mL/g substrate). A slight reduction in hydrophobicity of solid residues was observed upon pretreatment by EG: ChCl = 2:1 (2.12 mL/g substrate) and CA: ChCl = 1:1 (1.95 mL/g substrate), while CA: ChCl: EG = 1:1:2 afforded a significant decrease (0.07 mL/g substrate) due to removing the majority of hemicellulose and lignin. Furthermore, the introduction of EG in ternary DES protected the lignin structure to reduce the production of condensed lignin and impeded the excessive degradation of hemicellulose to decrease the formation of pseudo-lignin [45]. It was confirmed that the hydrophobicity of solid residues pretreated by ternary DESs was lower than that of binary DESs, which was mainly caused by the decrement content of condensed lignin and pseudo-lignin on the cellulose surface, in agreement with the previous literature [45]. The low hydrophobicity would decrease the non-productive adsorption between cellulase and lignin to increase the enzymatic hydrolysis efficiency [34].

3.2.6. Accessibility analysis of cellulose-rich residues

The cellulose accessibility of DES-pretreated solid residues was examined by the Direct Red 28 dye method (Table 2). It could be seen that the raw wheat straw had the lowest accessibility of 98.65 mg/g. The accessibility of binary DES-pretreated samples was increased to 105.83 mg/g (EG: ChCl = 2:1) and 140.26 mg/g (CA: ChCl = 1:1), respectively. In contrast, the ternary DES (CA: ChCl: EG = 1:1:2) pretreatment enhanced the accessibility to 393.17 mg/g, revealing about 3-fold higher than the raw sample. This could be interpreted by the protective action of EG on the lignin structure in ternary DES, which caused the negligible condensation of lignin and pseudo-lignin on the cellulose surface. On the other hand, an appropriate amount of CA could effectively remove hemicellulose and lignin, which was able to significantly increase cellulose accessibility and provide more adsorption sites for enzymes [45]. The improved BET surface area of cellulose was also observed over the other ternary DES (ChCl-OA-EG) [45]. The changes in cellulose accessibility of the pretreated residues are positively associated with enzymatic digestibility [35]. Thus, the solid residue after pretreatment of ternary DES (CA: ChCl: EG = 1:1:2) would possess more outstanding enzymatic hydrolysis saccharification.

3.2.7. Enzymatic hydrolysis of cellulose-rich residues

The enzymatic hydrolysis efficiency of cellulose is a crucial parameter to evaluate the effects of pretreatment [47]. After enzymatic hydrolysis for 72 h, the glucose yield of raw wheat straw was only 22.84%, while binary DESs pretreatment, including EG: ChCl = 2:1 and CA: ChCl = 1:1, could afford slightly increased glucose yield of 32.59% and 39.84%, respectively (Fig. 3a). The poor enzymatic hydrolysis of raw wheat straw could be due to the undisrupted linkages of LCC, which is in line with the previous research [16]. The inconspicuous improvement effect of binary DESs-pretreated substrates on enzymatic hydrolysis could be due to the absence of CA (insufficient acidity) or EG (high viscosity), resulting in inadequate removal of lignin and hemicellulose, which thus seriously decreased cellulose accessibility and cellulase adsorption [27]. Surprisingly, the glucose yields of different ternary DESs-pretreated substrates continuously increased from 45.60% to 98.73% (Fig. 3a and b). Especially, the glucose yield plateaued at 98.73% when wheat straw was pretreated by CA: ChCl: EG = 1:1:2, which was 4.32-fold higher than that of raw wheat straw. The suitable

content of CA (pKa: 3.13) (see section 3.1.1) in ternary DES provided sufficient protons (α-β: 0.78) (Table 2) to availably cleave the ether or ester bonds of LCC linkages, causing effective removal of hemicellulose and lignin. Meanwhile, an appropriate amount of EG introduction stabilized the structure of lignin and xylose to reduce the deposition of condensed lignin and pseudo-lignin on the cellulose surface, leading to an excellent enzymatic hydrolysis efficiency [19,45]. The optimal ternary DES (CA: ChCl: EG = 1:1:2) pretreated wheat straw obtained the highest glucose yield (98.73%), which would be more conducive to the subsequent fermentation. By contrast, relatively low glucose yields were obtained on the other ternary DESs, such as ChCl-EG-PTSA (78.40%) [48] and ChCl-OA-EG (85.72%) [45].

As illustrated in Fig. 3c, the enzymatic hydrolysis of wheat straw pretreated by CA: ChCl: EG = 1:1:2 at 100 °C displayed a higher glucose yield (98.73%) than that pretreated at 140 °C (86.53%), although the delignification (92.37%) of the former was lower than the latter (95.81%) (Table S1) [27]. Too high pretreatment temperature not only led to lignin condensation but also caused the carbohydrate to produce inhibitors and pseudo-lignin, which might be wrapped on the cellulose surface and result in non-productive adsorption of cellulase [31]. Also, excessive lignin removal could result in the collapse of pore morphology between cellulose to reduce cellulase accessibility, causing inferior enzymatic hydrolysis [27]. As shown in Fig. 3d, the glucose yield of the DESs-pretreated-sample progressively increased from 38.68% to 98.73% with prolonging the time from 3 to 12 h, then decreased to 87.25% after 24 h. A relatively longer time (12 h) resulted in more complete penetration of DES into the cell wall of wheat straw, improving the degradation of lignin and hemicellulose. However, the excessive extension of pretreatment time (24 h) caused superabundant removal of hemicellulose and lignin, leading to the formation of a dense structure between cellulose, which thus reduced cellulase accessibility [49]. Consequently, the superior enzymatic hydrolysis was observed upon pretreatment by CA: ChCl: EG = 1:1:2 at 100 °C for 12 h.

4. Characterization of extracted lignin

The functional group, molecular weight, polydispersity, purity, and chemical structure of extracted lignin significantly affected its utilization efficiency [38,50]. Given this, the obtained lignin was characterized by FT-IR, GPC, and NMR techniques.

4.1. Fourier transform infrared spectra analysis of extracted lignin

FT-IR analysis was used to assess the changes in functional groups of the DES-extracted lignin (Fig. S4). The characteristic bands at 1255, 1357, and 832 cm^{-1} were attributed to syringyl (S), guaiacyl (G), and *p*-hydroxyphenyl (H), respectively, inferring that the extracted lignin was the H-G-S lignin type [51]. The absorption signals at 2925 and 2860 cm^{-1} represented the alkane C-H stretching vibrations in $–CH_3$ and $–CH_2$- [43]. The peaks of aromatic skeletal vibration (1610, 1510, 1440 cm^{-1}) and C-O stretch in primary and secondary alcohols or aliphatic esters (1125, 1030 cm^{-1}) in extracted lignin were observed [43]. It was demonstrated that the structure of DES-extracted lignin was kept relatively intact due to the in-situ stabilization effect of EG on the lignin structure. Meanwhile, the characteristic signals of hemicellulose (e.g., 1730 cm^{-1}) and cellulose (e.g., 900, 1103 cm^{-1}) were not detected, proving that the obtained lignin had high purity because of the effective fractionation of CA on the hemicellulose and cellulose. Furthermore, the absorption peaks at 1710 and 1220 cm^{-1} disclosed that lignin extracted by DES was assigned to the C-OOHs, meaning that partial organic acid (e.g., CA and MA) was incorporated onto lignin fractions, which could reduce the cellulase non-productive adsorption [52]. Interestingly, the obtained lignin of ternary DESs pretreatment (e.g., CA: ChCl: EG = 1:1:2) showed a stronger intensity of the signals (1710 and 1220 cm^{-1}) than those of binary counterparts, implying that the lignin extracted by ternary DES had lower non-productive adsorption.

4.2. Nuclear magnetic resonance analysis of extracted lignin

The lignin extracted by CA: ChCl: EG = 1:1:2 pretreatment of wheat straw was determined by the NMR technique (Fig. S5). The proton absorption bands of the aliphatic acetates (1.4 and 2.2 ppm), aromatic acetates (2.2–2.5 ppm), methoxy (3.5–4.1 ppm), and aromatic ring (6.5 ppm) in the DES-extracted lignin were detected. Moreover, the carbons peaks of the aliphatic side chain region (20–40 ppm) and methoxy (54–57 ppm), and ester group (175–180 ppm) in the extracted lignin were also observed [43]. It meant that a large number of functional groups belonging to the lignin structure were retained and existed in the DES-extracted lignin, illustrating that the lignin structure was well protected. Surprisingly, the disappearance of ^{1}H NMR signals at 6.35–6.65 ppm and ^{13}C NMR signals at 106–109 ppm in aromatic regions were assigned to the condensed lignin, signifying the absence of lignin condensation [53,54]. Interestingly, ^{1}H NMR peaks at 5.17 ppm and ^{13}C NMR peaks at 72 and 85.9 ppm assigned to β-O-4 linkages were not detected, whereas the ^{1}H NMR bands at 3.56, 3.71 and 4.8 ppm, as well as ^{13}C NMR bands at 61.6 and 70.1 ppm corresponding to the protected-β-O-4, were clearly observed, further implying that the lignin structure was protected successfully [43,50]. It can be concluded that the presence of CA could destroy the β-O-4 linkages while EG could stabilize the reactive intermediates to produce protected-β-O-4 in ternary DES pretreatment (Fig. 4). Besides, the absence of carbohydrates peaks at 90 and 102 ppm in ^{13}C NMR and 4.0–5.5 ppm in ^{1}H NMR spectra illustrated that nearly all carbohydrates were removed (Table 1) and the extracted lignin possessed high purity (Table 3) [38]. The results manifested that lignin was selectively cleaved and directionally protected under the synergistic effect of CA and EG in ternary DES.

4.3. Molecular weight distribution and purity analysis of extracted lignin

The weight-average (M_w), number-average (M_n) molecular weight, and polydispersity index (PDI, M_w/M_n) of extracted lignin were evaluated by GPC (Table 3). The PDI of lignin extracted by EG: ChCl = 2:1 and CA: ChCl = 1:1 was 1.701 (M_w: 4929 g/mol, M_n: 2898 g/mol) and 1.672 (M_w: 4103 g/mol, M_n: 2454 g/mol), respectively. By comparison, the lignin extracted by ternary DES (CA: ChCl: EG = 1:1:2) had a relatively lower PDI of 1.202 (M_w: 2742 g/mol, M_n: 2281 g/mol). The inferior PDI could be ascribed to the effective cleavage of CA on the β-O-4 linkages, while superior protection of EG on the activated intermediates [36]. The results indicated that the lignin extracted by ternary DES was homogenous polymers and kept relatively complete without condensation reaction [55]. With respect to the reported method for extraction of lignin by ternary DES (ChCl/AA/Gly) pretreatment (M_w: 3962 g/mol, M_n: 2534 g/mol, PDI: 1.563) [56], the obtained lignin in this work had greater potential in the production of high-value chemicals, in terms of relatively lower PDI, M_w and M_n. Also, the lignin extracted by CA: ChCl: EG = 1:1:2 pretreatment had a high purity of 96.6% (Table 3), verifying the unexceptionable delignification efficiency. In short, the ternary DES-extracted lignin has broad prospects for industrial production with less condensed, low polydispersity, and high purity.

4.4. Mechanism of fractionation and protection of lignin

4.4.1. Mechanism of lignin fractionation

FT-IR results of DES (Fig. S6) confirmed the formation of intermolecular hydrogen bonds between HBA and HBDs, considering that the absorption signals of O-H at about 3300 cm^{-1} in DESs became broader than those in pure EG and organic acid [12]. The K-T parameters of DES, such as α (acidity: hydrogen-bond-donating ability), β (basicity: hydrogen-bond-accepting ability), and π* (polarizability), can forecast the fractionation mechanism since the proton-catalyzed bond breakage is the primary driving force for DES pretreatment [43]. In detail, the relatively high α - β value of DES exhibits a high net hydrogen-donating ability and forms strong hydrogen bonds with lignin, which can promote

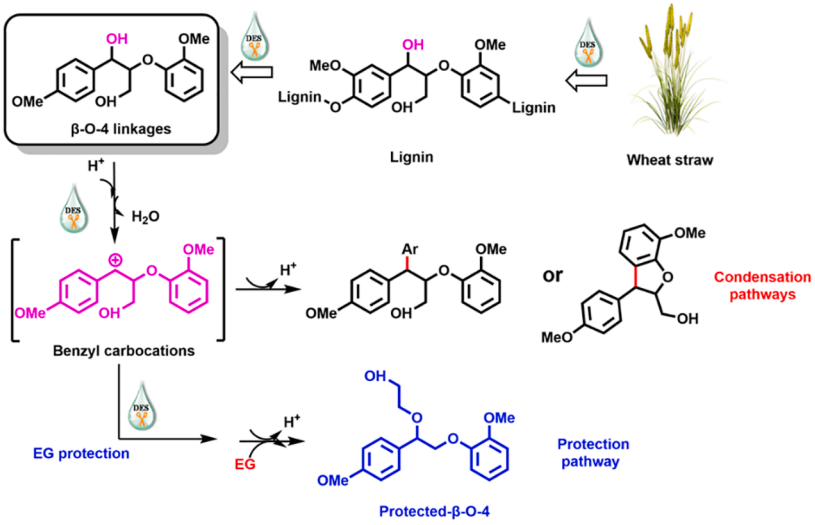


Fig. 4. The pathways of lignin condensation and protection.

Table 3
The M_w, M_n, PDI, and purity of various extracted lignin.

Entry	DES	M_w	M_n	PDI	Purity
1	EG: ChCl = 2:1	4929	2898	1.701 ± 0.09	83.8 ± 2.12
2	CA: ChCl = 1:1	4103	2454	1.672 ± 0.07	85.1 ± 1.34
3	CA: ChCl: EG = 5:1:2	4828	3082	1.567 ± 0.05	93.7 ± 1.56
4	CA: ChCl: EG = 2:1:2	4088	3191	1.281 ± 0.10	95.8 ± 2.19
5	CA: ChCl: EG = 1:1:2	2742	2281	1.202 ± 0.02	96.6 ± 1.07
6	CA: ChCl: EG = 1:2:2	3842	2992	1.284 ± 0.11	95.9 ± 2.31
7	CA: ChCl: EG = 1:5:2	3811	2830	1.347 ± 0.10	95.5 ± 1.25
8	CA: ChCl: EG = 1:1:4	3655	2649	1.380 ± 0.08	94.9 ± 2.87
9	CA: ChCl: EG = 1:1:8	4634	3206	1.445 ± 0.06	94.1 ± 1.76
10	FA: ChCl: EG = 1:1:2	3807	2958	1.287 ± 0.01	95.6 ± 2.14
11	AA: ChCl: EG = 1:1:2	5589	3645	1.533 ± 0.03	94.5 ± 1.52
12	LA: ChCl: EG = 1:1:2	3845	2840	1.354 ± 0.07	94.8 ± 2.08
13	MA: ChCl: EG = 1:1:2	3594	2844	1.264 ± 0.05	95.4 ± 1.54
14	BA: ChCl: EG = 1:1:2	5353	3384	1.582 ± 0.04	92.3 ± 2.46

lignin degradation [25]. A high π* value indicates the strong fractionation and solubilization capacity of lignin due to the relatively high polarity [24].

As shown in Table 2, the ternary DES (CA: ChCl: EG = 1:1:2) possessed the highest values of α-β (0.78) and π* (1.19), resulting in superior lignin removal (92.37%) (Table 1). However, binary DESs (EG: ChCl = 2:1 and CA: ChCl = 1:1) with lower values of α-β (0.14 and 0.32) and π* (0.81 and 0.89) exhibited inferior delignification (35.67% and 41.52%) due to lower net hydrogen donating. It was evidenced that a suitable amount of CA in the ternary DES could effectively break the ether or ester bonds of LCC linkages because of its high net hydrogen-donating ability, while an appropriate content of EG in the ternary DES could reduce the viscosity to promote mass transfer efficiency. Furthermore, the Cl^- in ChCl can form hydrogen bonds with –OH in lignin, facilitating lignin fractionation [57]. Accordingly, the ternary DES had remarkable fractionation efficiency, resulting in excellent enzymatic hydrolysis efficiency.

4.4.2. *Mechanism of lignin protection in ternary deep eutectic solvent (CA/ChCl/EG)*

According to the results of this work and the reported investigations, the mechanism of lignin protection of ternary DES (CA/ChCl/EG) pretreatment of wheat straw was proposed. The ether or ester bonds in LCC linkages were cleaved by the activated protons from CA, leading to the solubilization of hemicellulose and lignin [58]. In Fig. 4, the β-O-4 linkages of lignin were easy to protonate by protons and dehydrate to produce benzyl carbocations. The undesired lignin condensation occurred via benzyl carbocations reacting with aromatic rings, resulting in the formation of recalcitrant C-C bonds and the loss of valuable β-O-4 linkages [50]. Interestingly, the incorporation of EG could stabilize the reactive intermediates by trapping the formed benzyl carbocations, thereby inhibiting the lignin condensation [41,59], which had been confirmed by FT-IR (Fig. S4) and NMR spectra (Fig. S5). In general, CA: ChCl: EG = 1:1:2 pretreated wheat straw exhibited excellent fractionation efficiency and in-situ lignin protection ability due to the suitable ratio of CA, EG, and ChCl in the ternary DES.

4.5. *Lipid production from wheat straw pretreated by the ternary deep eutectic solvent*

4.5.1. *Fermentation of deep eutectic solvent waste liquid by Trichosporon cutaneum*

The undetoxified DES waste liquids were diluted four times and utilized as carbon sources for lipid production. As shown in Fig. 5a, b, and c, in the first 24 h, the yeast growth was almost not observed on the binary DESs (EG: ChCl = 2:1 and CA: ChCl = 1:1) waste liquids, while obviously increased on the ternary DES (CA: ChCl: EG = 1:1:2). It could be illustrated that the stabilization of EG on the xylose structure in ternary DES resulted in lower content of inhibitors (Table 1). The reduction of inhibitors was not only beneficial to the growth of microorganisms but also decreased the cost of detoxification and the loss of total sugars, which would lead to superior fermentation efficiency [60]. Surprisingly, the glucose and xylose were almost consumed entirely, verifying that *T. cutaneum* can simultaneously transfer glucose and xylose to lipids [61].

After fermentation for 144 h, *T. cutaneum* achieved 5.37 and 5.76 g/L of DCW, 0.59 and 0.73 g/L of lipid yield, 11.01 and 12.62% of lipid content from EG: ChCl = 2:1 and CA: ChCl = 1:1 waste liquid, respectively, while afforded 6.36 g/L DCW, 1.22 g/L lipid yield, and 19.12% lipid content from CA: ChCl: EG = 1:1:2 waste liquid, respectively. It was observed that the existence of CA in ternary DES facilitated the degradation of hemicellulose into more xylose and in parallel EG stabilized the xylose to decrease the formation of toxic inhibitors [19]. Satisfactorily, another ternary DES waste liquids also revealed high lipid yields (0.84–1.00 g/L) and lipid contents (15.21–17.17%) (Fig. 5d), manifesting the wide application of DES waste liquids by *T. cutaneum* in lignocellulose fermentation. Moreover, the optimum ternary DES (CA: ChCl: EG = 1:1:2) waste liquids resulted in higher lipid yield (1.22 *vs* 0.89 g/L) than the fermentation of spent wash hydrolysate by *T. cutaneum* [62]. Importantly, the DES waste liquids as the substrates were not required to be detoxified, which greatly reduced the cost of

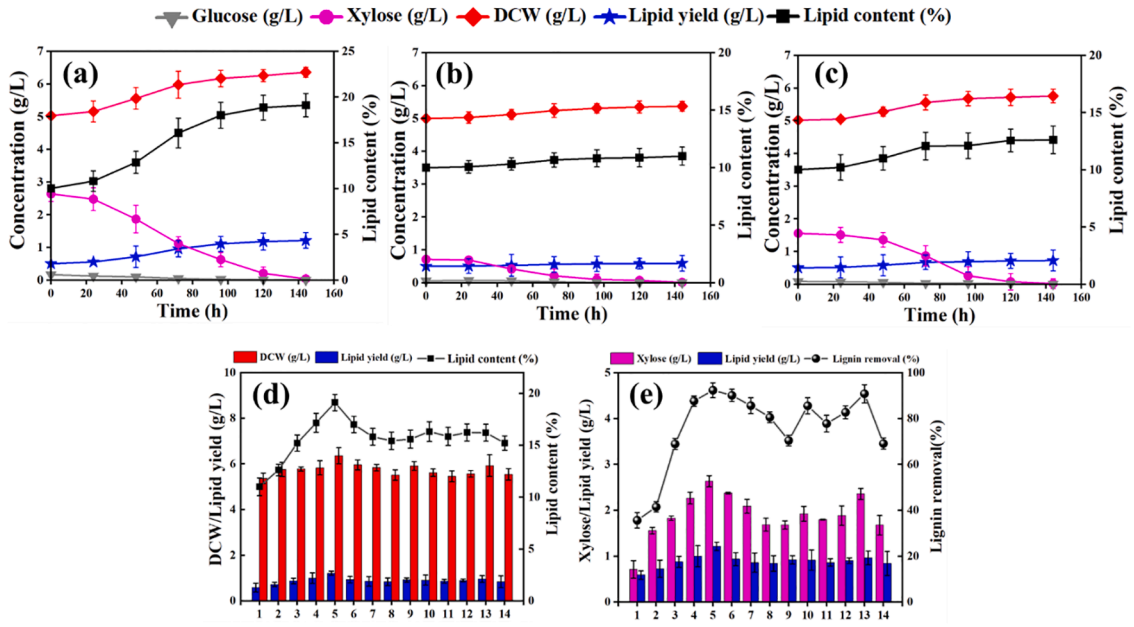


Fig. 5. Profiles of glucose, xylose, DCW, lipid yield, and lipid content using *T. cutaneum* fermented on different DES waste liquids: (a) CA: ChCl: EG = 1:1:2, (b) EG: ChCl = 2:1, (c) CA: ChCl = 1:1, (d) NO.1–14 DESs, and (e) the relationship among lignin removal and xylose yield and lipid yield.

fermentation compared to the other spent wash. Herein, the ternary DES waste liquid contained a high concentration of sugars but trace amounts of inhibitors (Table 1), possessing superior competition to be used as the substrate for lipid production.

Fig. 5e showed that with the increase of lignin removal (35.67–92.37%), xylose yield (0.71–2.64 g/L) and lipid yield (0.59–1.22 g/L) were enhanced increasingly. DES pretreatment could availably destroy the β-O-4 linkages to remove more hemicellulose and lignin, and then hemicellulose was further hydrolyzed to give more xylose (Table 1), thereby increasing lipid yield. In brief, the increased efficiency of DES pretreatment could promote fermentation efficiency.

4.5.2. *Fermentation of cellulose-rich hydrolysates by Trichosporon cutaneum*

The cellulose-rich hydrolysates were also used as carbon sources for lipid production by *T. cutaneum*. In Fig. 6, the yeast growth started in the first 12 h without an obvious delay period, and the fermentation cycle (72 *vs* 144 h) was significantly shorter than that of DES waste liquids (Fig. 5), which was primarily due to the absence of inhibitors [21]. As listed in Fig. 6a, b, and c, DCW, lipid yield, and lipid content of residue hydrolysates pretreated with CA: ChCl: EG = 1:1:2 was 30.72 g/L, 12.68 g/L, and 41.26%, respectively, which were remarkably higher than those of EG: ChCl = 2:1 (DCW: 11.65 g/L, lipid yield: 2.54 g/L, and lipid content: 21.81%) and CA: ChCl = 1:1 (DCW: 14.79 g/L, lipid yield: 4.02 g/L, and lipid content: 27.19%). Moreover, *T. cutaneum* also obtained relatively higher lipid yields (4.14–10.73 g/L) and lipid contents (25.68–38.69%) than other ternary DES-pretreated residue hydrolysates (Fig. 6d). The protective function of EG on lignin and xylose reduced the deposition of condensed lignin and pseudo-lignin on the cellulose surface and improved the enzymatic hydrolysis to release a high

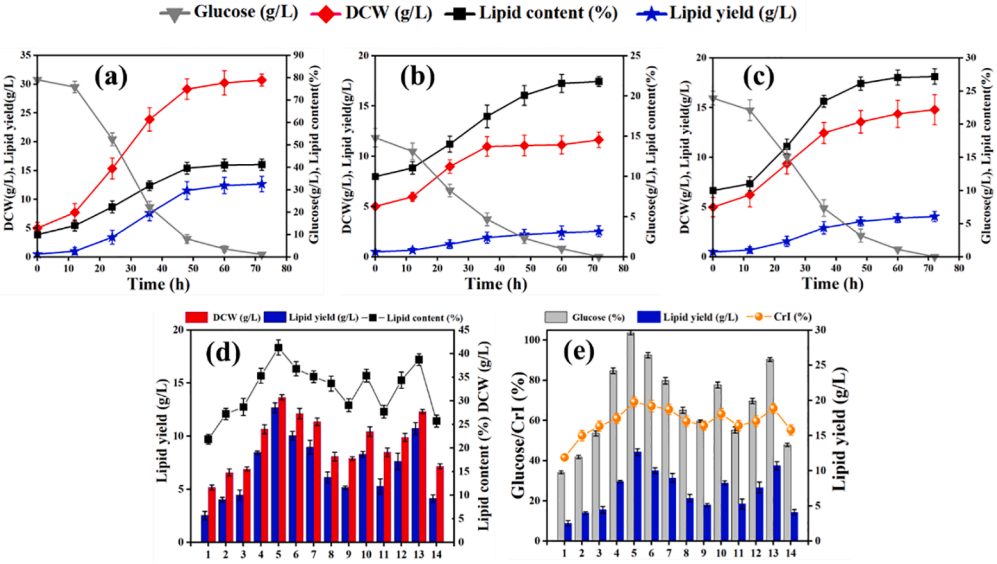


Fig. 6. Profiles of glucose, DCW, lipid yield, and lipid content using *T. cutaneum* fermented on different residue hydrolysates: (a) CA: ChCl: EG = 1:1:2, (b) EG: ChCl = 2: 1, (c) CA: ChCl = 1:1, (d) NO.1–14 DESs, and (e) the relationship among crystallinity index (CrI), glucose yield and lipid yield.

concentration of glucose, which led to better fermentation efficiency of the ternary DES than that of the binary DESs [19]. As expected, *T. cutaneum* cultivated on cellulose-rich hydrolysates pretreated with DES almost entirely consumed glucose. In addition, the solid residue hydrolysates as substrates achieved considerably higher lipid yields (2.54–12.68 g/L) than that of DES waste liquids (0.59–1.22 g/L) because the glucose was the only carbon source with the absence of any inhibitors [21]. Indeed, the solid residue pretreated by ternary DES (CA: ChCl: EG = 1:1:2) was beneficial to the production of lipids from *T. cutaneum*.

The results of lipid yield/content obtained using various pretreatment and fermentation methods were summarized in Table 4. The observed lipid yield and lipid content on wheat straw residue hydrolysate by *T. cutaneum* (12.68 g/L and 41.26%) were higher than those on corn stover hydrolysate by *T.cutaneum* (3.1 g/L and 23.5%) [29], arundo donax L. hydrolysate by *Lipomyces starkeyi* (7.5 g/L and 34%) [21], and corncob hydrolysate by *T.cutaneum* (12.3 g/L and 32.1%) [63]. The findings of the above investigations proved the high quality of wheat straw and further emphasized its good potential in lipid production. The ternary DES-enabled in-situ lignin protection strategy exhibited superior pretreatment efficiency, resulting in outstanding fermentation efficiency.

In Fig. 6e, with the increase in the CrI (32.57–69.24%), the glucose yield (32.59–98.73%) and lipid yield (2.54–12.68 g/L) progressively increased. The removal of hemicellulose and lignin after DESs pretreatment resulted in the exposure of cellulose and the increase of CrI. High CrI illustrated that the cellulose was completely exposed and the cellulose surface was covered with less content of condensed lignin and pseudo-lignin, which promoted cellulose accessibility and caused high glucose yield and lipid yield. The increased CrI of cellulose pretreated by the ternary DES had positive effects on the glucose yield in the enzymatic process and the lipid yield in fermentation.

4.6. Fatty acid methyl esters composition analysis

The distribution of fatty acids in microbial lipids is a vital factor influencing the quality of subsequent products. The main composition of fatty acids was palmitic acid (C16:0), stearic (C18:0), oleic (C18:1), linoleic (C18:2), and linolenic acids (C18:3), which accounted for 92.72–97.27% and 94.42–98.44% of the total lipids obtained from DES waste liquids (Fig. 7a) and residue hydrolysates (Fig. 7b), respectively [1,63]. The obtained results showed that *T.cutaneum* achieved higher content of C18:0 (26.49%, 27.51%) in binary DESs (EG: ChCl = 2:1, CA: ChCl = 1:1) waste liquids than that of C18:0 (6.42%) in ternary DES (CA: ChCl: EG = 1:1:2), whereas the binary DESs afforded lower content of C16:0 (28.62% and 27.81% *vs* 36.24%), C18:1 (29.98% and 31.62% *vs* 49.04%), and C18:2 (4.31% and 4.75% *vs* 5.24%) than the ternary DES, respectively. The absence of inhibitors in ternary DES might contribute to its higher unsaturated fatty acids (54.61%) than that of binary DESs (37.61%, 38.98%) [64]. It was worth noting that the higher unsaturated fatty acids are more favorable to the uniform combustion of diesel fuel and keep higher thermal power [56]. In addition, the lipid-derived FAMEs from binary and ternary DESs residue hydrolysates possessed high percentage of unsaturated fatty acids (~55.90%) due to the absence of inhibitors, which was similar to the composition of typical biodiesel [21]. Overall, wheat straw pretreated with ternary DES (CA: ChCl: EG = 1:1:2) was an appropriate feedstock for biodiesel production.

4.7. Mass balance

The mass balance flow diagrams of three DESs pretreatment of wheat straw were shown in Fig. 8. Based on the binary DESs pretreatment with EG: ChCl = 2:1 and CA: ChCl = 1:1, 2.5 g and 3.8 g lipids were obtained from 100.0 g raw wheat straw by *T. cutaneum*, respectively. Surprisingly, 100.00 g raw wheat straw undergoing ternary DES (CA: ChCl: EG = 1:1:2) pretreatment produced 8.7 g lipids, in which 3.6 g lipids derived from hemicellulose fraction while 5.1 g from cellulose hydrolysate. It could be found that EG in ternary DES not only protected the lignin structure and caused the decreased deposition of condensed lignin on the cellulose surface to enhance the release of glucose in cellulose hydrolysate, but also stabilized the xylose and resulted in the decrement of pseudo-lignin to promote the release of xylose in hemicellulose fraction. Thus, the highest release of total fermentation sugars was achieved from wheat straw under the pretreatment of CA: ChCl: EG = 1:1:2, leading to an excellent performance of lipid production. In brief, the ternary DES pretreatment had potential application value and innovation in the production of lipids from raw lignocellulosic biomass by *T. cutaneum*.

4.8. Techno-economic analysis

A techno-economic analysis is usually utilized for estimating the economic feasibility of biorefinery [65]. According to previous studies [65–67], lower temperatures, higher content of fermentable sugars, lower enzyme dosage, trace amount of toxic inhibitors, value-added products, cheaper raw materials, and a simple process of pretreatment could lead to higher energy efficiency and economic benefits. In this work, the relatively lower temperature (100 °C), lower enzyme dosage (25 FPU/g) with higher glucose yield (98.73%), almost complete recovery of xylose (10.55 g/L) from hemicellulose, and higher purity (96.6%) of lignin without condensation were achieved, which led to the reduction of production cost [65]. Further, the green and economic components of the ternary DES, cheap wheat straw, and the simple pretreatment process also resulted in higher economic benefits [50,66]. Another advantage of this work process was the production of microbial lipid from DES waste liquids without detoxification, significantly saving the total cost of biodiesel production [10]. Overall, the strategy had the commercial viability potential to upgrade the lignocellulose for biodiesel production.

5. Conclusions

A novel ternary DES-enabled in-situ lignin protection strategy could protect the lignin structure, impede the excessive degradation of hemicellulose but to fermentable sugars (e.g., xylose: 10.55 g/L) and reduce the formation of fermentable inhibitors, which improve the utilization efficiency of hemicellulose. The mechanisms of both lignin fractionation and the in-situ lignin protection during pretreatment enabled by ternary DES were systematically illustrated. Furthermore, a high yield of

Table 4
Comparison of lipid yield/content obtained using various pretreatment and fermentation methods.

Microorganism	Raw material	Pretreatment method	DCW (g/L)	Lipid yield (g/L)	Lipid content (%)	Ref.
Lipomyces starkeyi MTCC 9519	Arundo donax L. hydrolysate	Microwave-assisted $FeCl_3$	22	7.5	34	[21]
Trichosporon cutaneum No. 2527	Corn stover hydrolysate	Dilute acid	15.4	3.1	23.5	[29]
Trichosporon cutaneum CX1	Corn stover hydrolysate	Dilute acid	10.2	3.1	30.4	[29]
Trichosporon cutaneum ACCC 20271	Corncob hydrolysate	Enzymatic hydrolysis	38.4	12.3	32.1	[62]
Trichosporon cutaneum AS 2.571	Corn stover hydrolysate	Dilute acid	19.3	7.6	39.2	[63]
Trichosporon fermentans CICC1368	Rice hull hydrolysate	Dilute acid	28.6	11.5	40.1	[64]
Trichosporon cutaneum ACCC 20271	Corn stover hydrolysate	Dilute acid	28	8.1	28.9	[65]
Trichosporon cutaneum ACCC 2019	Wheat straw hydrolysate	Ternary DES	30.7	12.7	41.3	This work

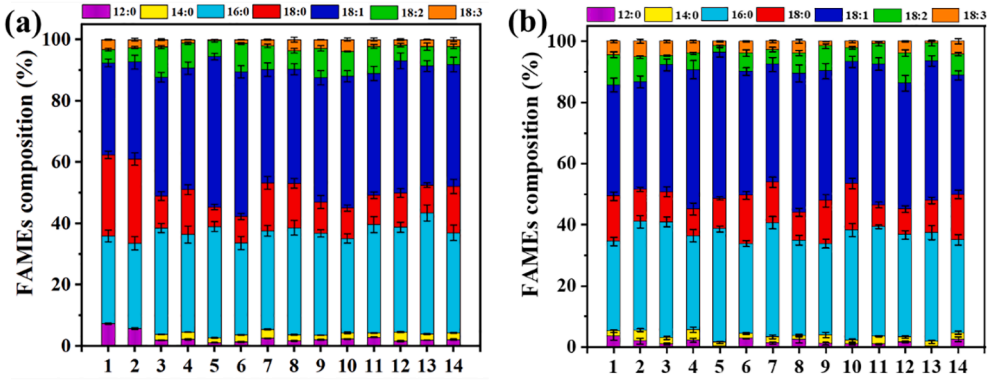


Fig. 7. The composition of lipid-derived FAMEs by *T. Cutaneum* from DES waste liquids (a), and cellulose-rich residue hydrolysates (b).

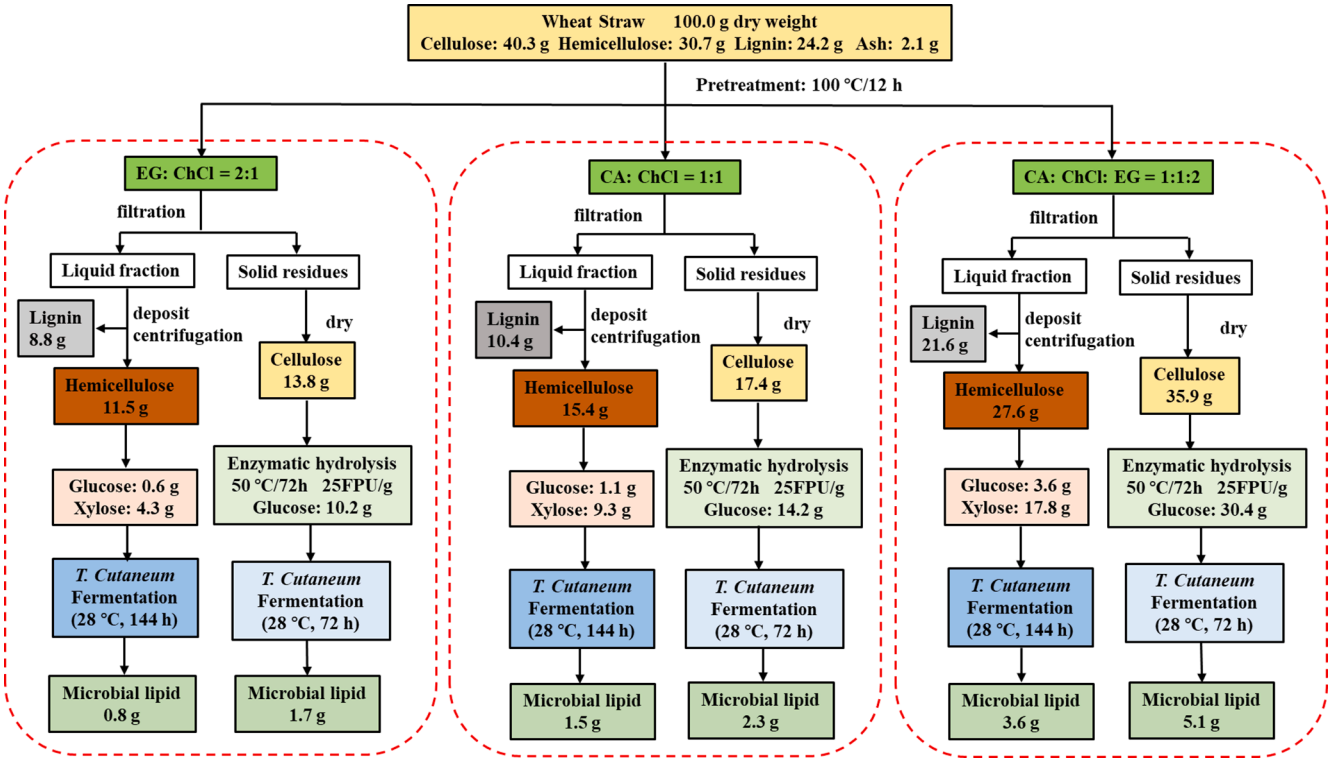


Fig. 8. Flow diagrams and mass balance of different DESs pretreatment for the production of microbial lipid from wheat straw.

microbial lipid (8.7 g/100 g) could be obtained from wheat straw by *T. cutaneum*, demonstrating superior pretreatment efficiency enabled by the ternary DES. The in-situ lignin protection and fractionation strategy provides a new avenue for lignocellulose pretreatment and further upgrading.

CRediT authorship contribution statement

Jinyu Tan: Methodology, Data curation, Investigation, Visualization, Writing – original draft. **Dayong Yu:** Methodology, Formal analysis. **Junfa Yuan:** Formal analysis. **Hongguo Wu:** Writing – review & editing. **Hangyu Luo:** Formal analysis. **Heng Zhang:** Formal analysis. **Xiaobing Li:** Formal analysis. **Hu Li:** Conceptualization, Funding acquisition, Project administration, Writing – review & editing. **Song Yang:** Conceptualization, Supervision, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.

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